

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy $< 90\%$ $\rightarrow$ system fails
- If latency > 70 ms $\rightarrow$ unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy $< 95\%$ $\rightarrow$ defects may be missed
- If processing time > 50 ms $\rightarrow$ production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, processing time, and computational requirements.